pfi

Python for Fixed Income

COURSE 4 · MACHINE LEARNING

PCA and t-SNE

WEEK 3 · DAY 4

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From numbers you chose to numbers the data choose

Course 3

Shapes you chose: 2s10s, 5s30s, the 2s5s10s fly. Weights a desk picked.

Day 11

Tenors so correlated a regression could hardly tell them apart.

Today: PCA

A few weighted sums that carry most of how the curve moved. No target.

Real Treasury data, January 2015 to October 2026. A component is a shape of movement in that history; a score is how much of that shape a past date had.

No component says what rates will do, or what anyone should do. Nothing after the last row.

The title says what the method does

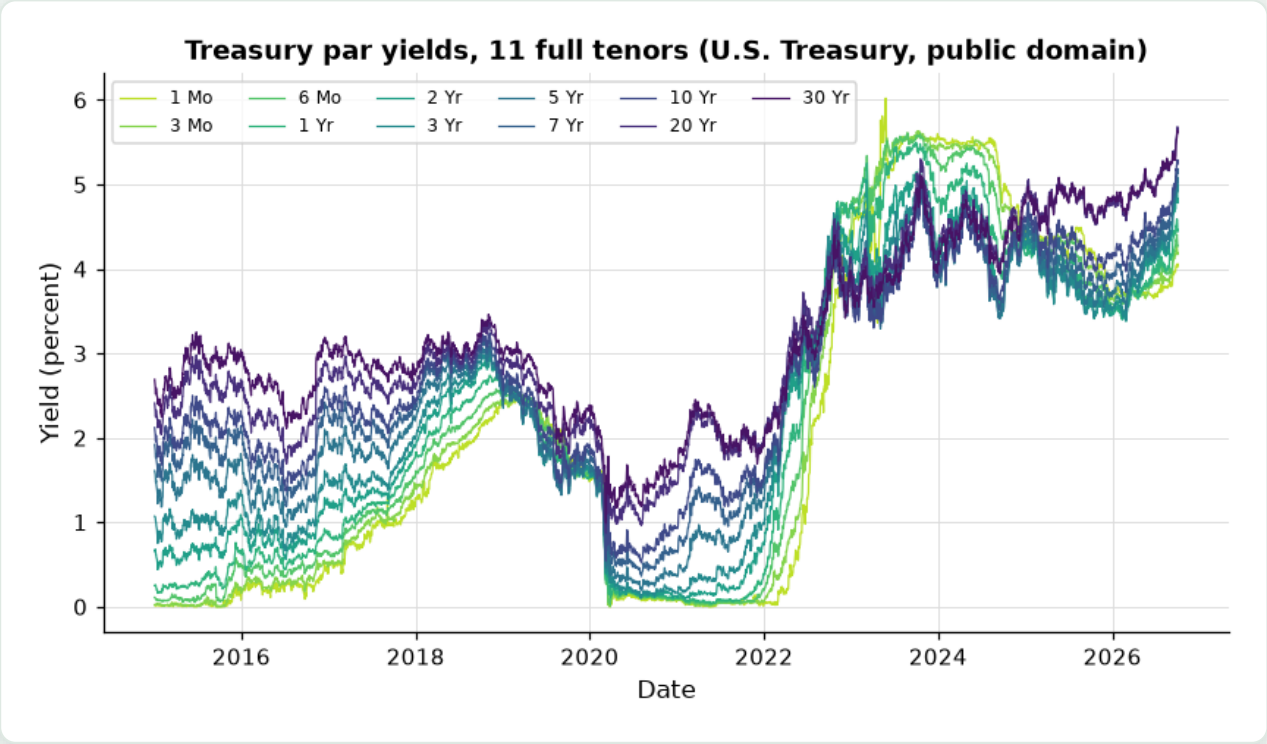
Karl Pearson, November 1901

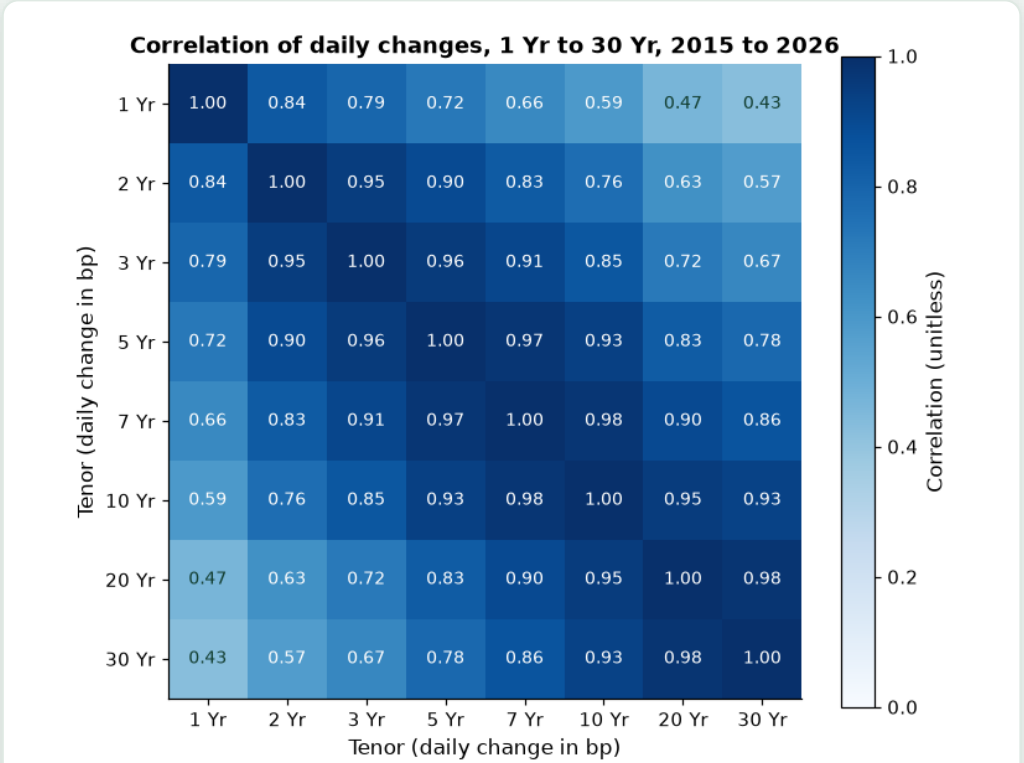
"On lines and planes of closest fit to systems of points in space." The Philosophical Magazine, volume 2, issue 11, pages 559 to 572.

Today the points are 2,940 market dates, each a point in eleven dimensions, one per tenor.

Source, read 4 October 2026: the Crossref record of the paper (api.crossref.org), DOI 10.1080/14786440109462720. Bibliographic facts only; the paper itself was not read for this course.

Many columns that move together



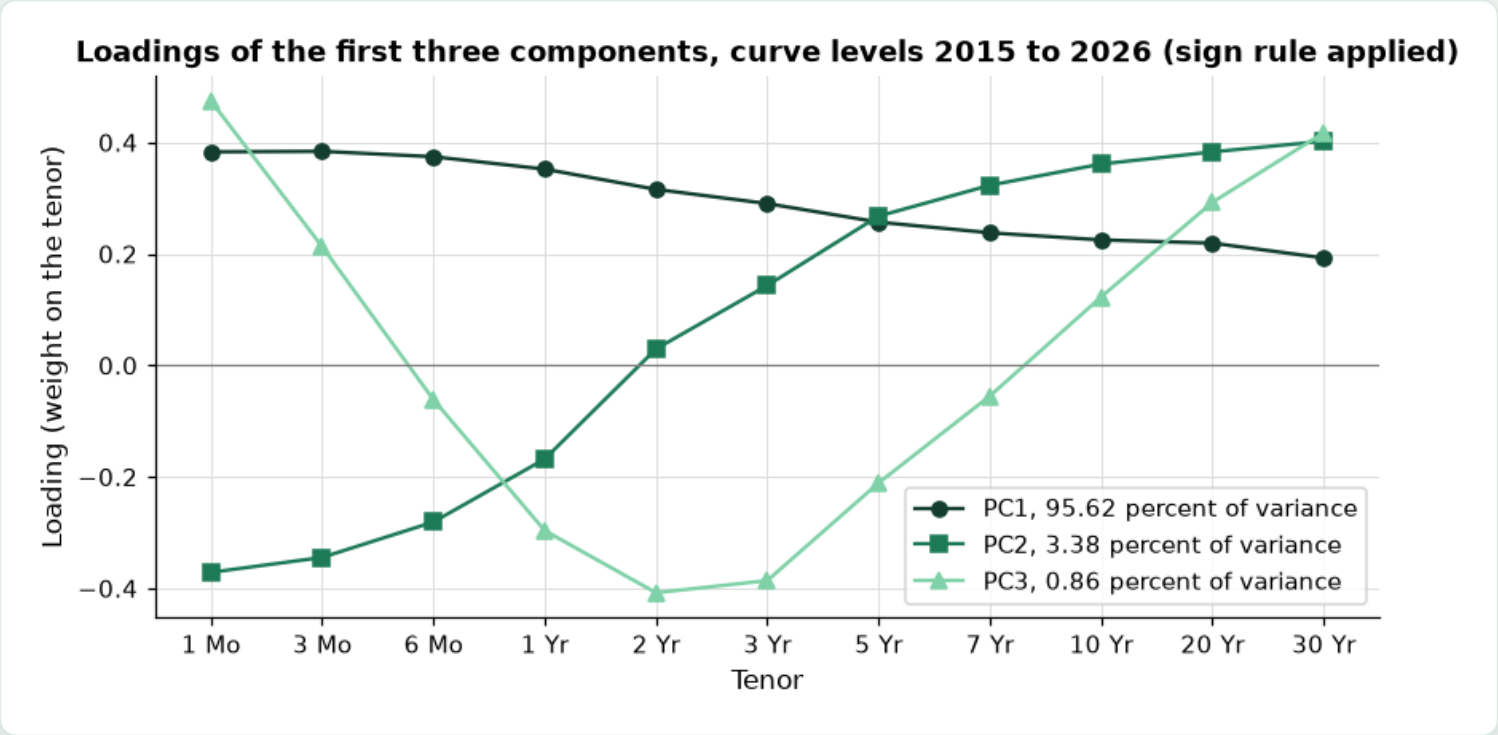


2,939 daily changes: 20 Yr with 30 Yr 0.98, 1 Yr with 30 Yr 0.43. Correlated columns carry less than their count.

How PCA works

The problem	Many columns move together; a few new ones can carry most of the movement
Components	The first direction spreads the dates out the most; the next, at right angles, the most of what is left
Loadings, scores	Loadings are a component's weights on the tenors; a date's score is its yields less averages, times loadings, summed
Variance	Each component carries a share, falling from one to the next
You control	How many components to keep, and whether to scale first

PCA on levels: loadings and their shapes

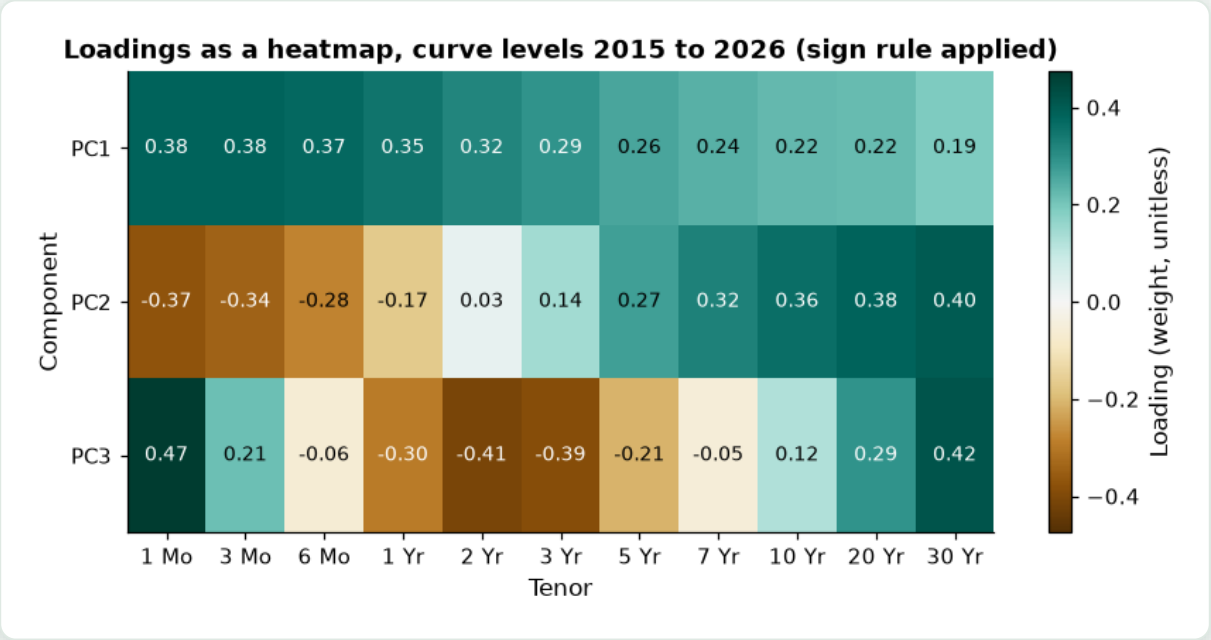


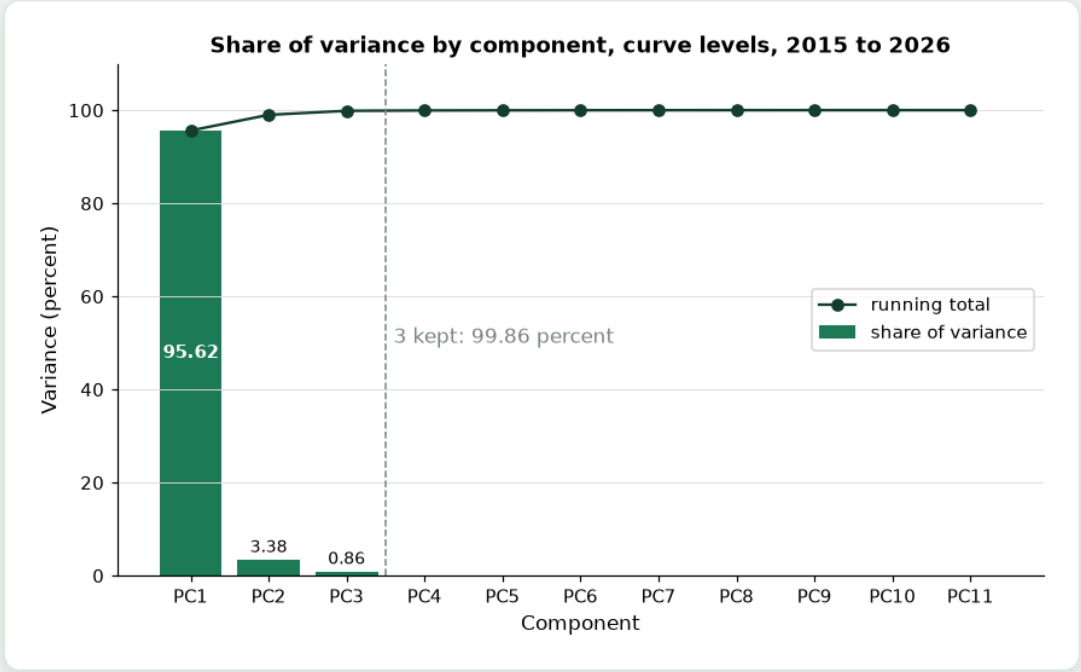
	Share	Shape
PC1	95.62%	level
PC2	3.38%	slope
PC3	0.86%	curvature

Three components: 99.86 percent of the variance of eleven yields.

Level, slope, curvature: names for shapes of weights in this history.

Two more ways to read a PCA

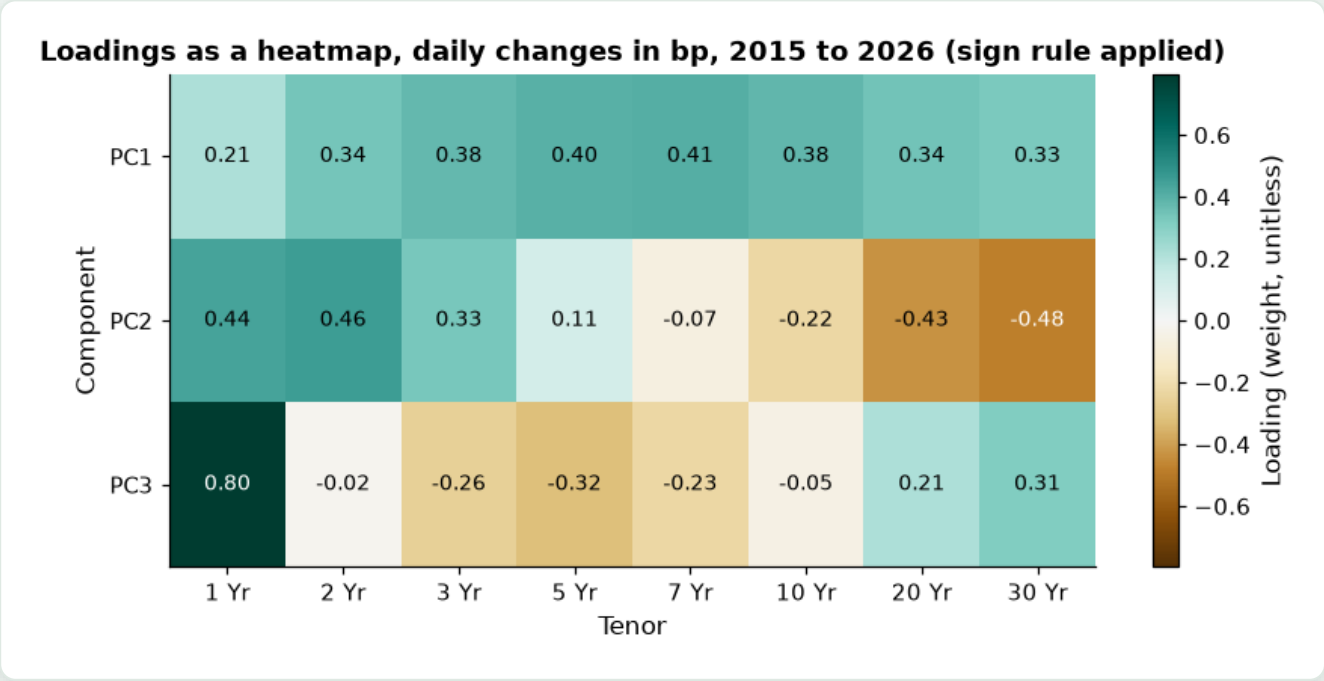




Heatmap: read along a row, colour is the sign.

Scree chart: bars per component, the running total as a line.

PCA on daily changes



Score	Course 3 measure	Corr
PC1	average change	0.9995
PC2	change in 2s10s	-0.9174
PC3	change in 2s5s10s	-0.4509

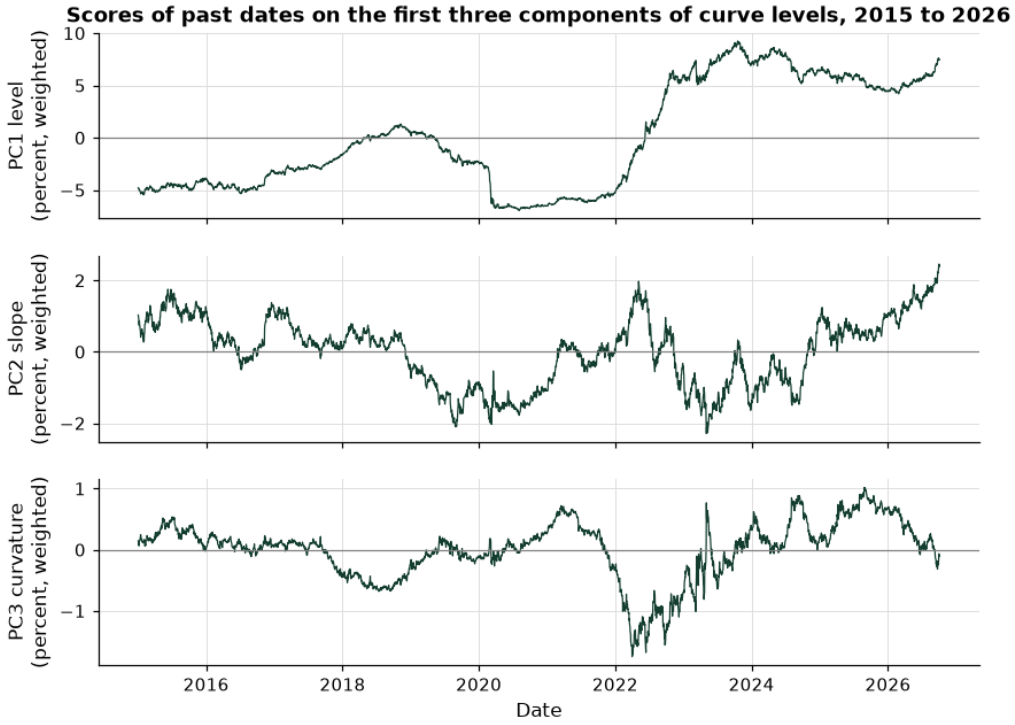
Shares: 85.04, 11.02, 2.11 percent (98.17 in three).

PC3 is mostly the 1 Yr (loading 0.796). The loadings, not the name, say what it is.

```
signs = np.where(loadings.sum(axis=0) < 0, -1.0, 1.0)
```

The sign rule: loadings of each component sum to a positive number.

Scores over time, as history



Score	Lowest	Highest
Level	2020-08-04	2023-10-19
Slope	2023-05-04	2026-09-30

Allowed names

level score

slope score

curvature score

Never called

a forecast

a view on rates

a signal

Each one a score of a past date, inside the file's span.

PCA as a feature step, and lookahead

The no-lookahead check, t = 2021-01-04

Fitted up to t, later rows removed: unchanged, True.

Fitted on the whole history: the same scores? False.

```
Pipeline([("pca", PCA(n_components=3)),
          ("line", LinearRegression())])
```

0.9 bp

three components, then a line

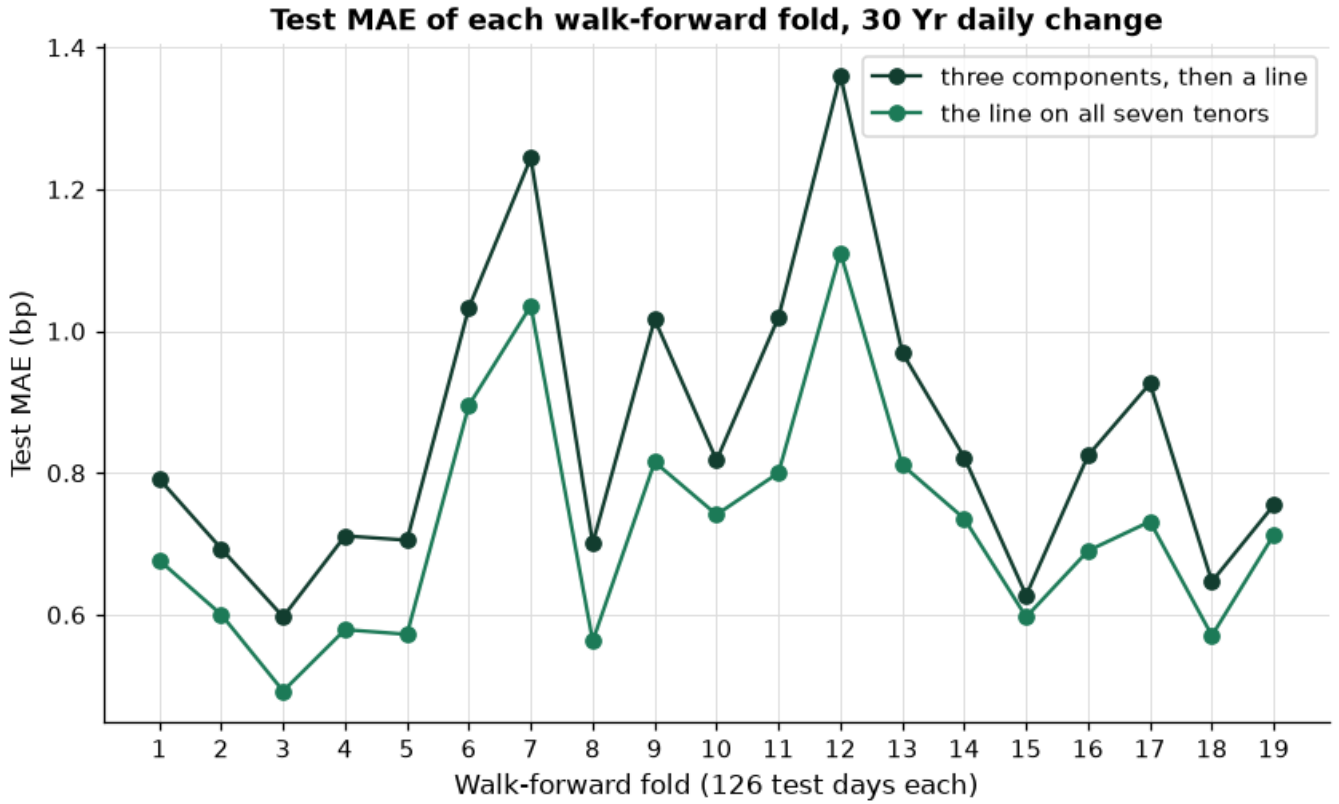
0.7 bp

the line on all seven tenors

Walk-forward, 19 folds, 2,394 test days: the 30 Yr change given the same day's other changes.

When PCA builds features it goes inside a pipeline, refitted on each fold's training rows.

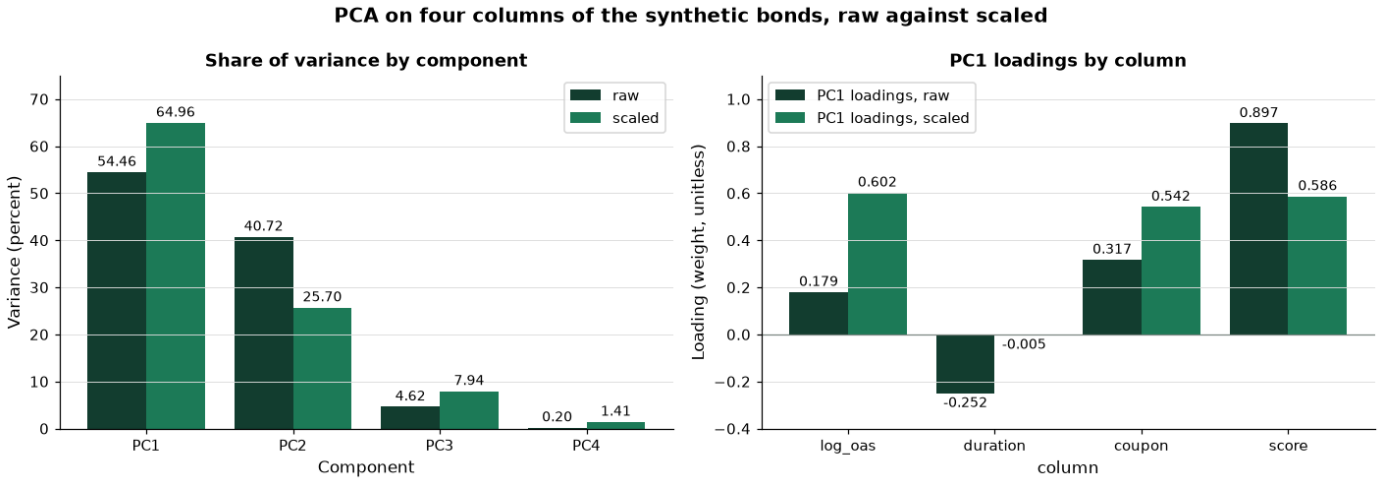
Two models, fold by fold



Pooled	R-squared	MAE	RMSE
3 components	0.9465	0.9 bp	1.2 bp
7 tenors	0.9607	0.7 bp	1.0 bp

A summary keeps most of the variance, not all of the detail. PCA describes and compresses; only sometimes is it better features.

Scaling before PCA, on mixed units

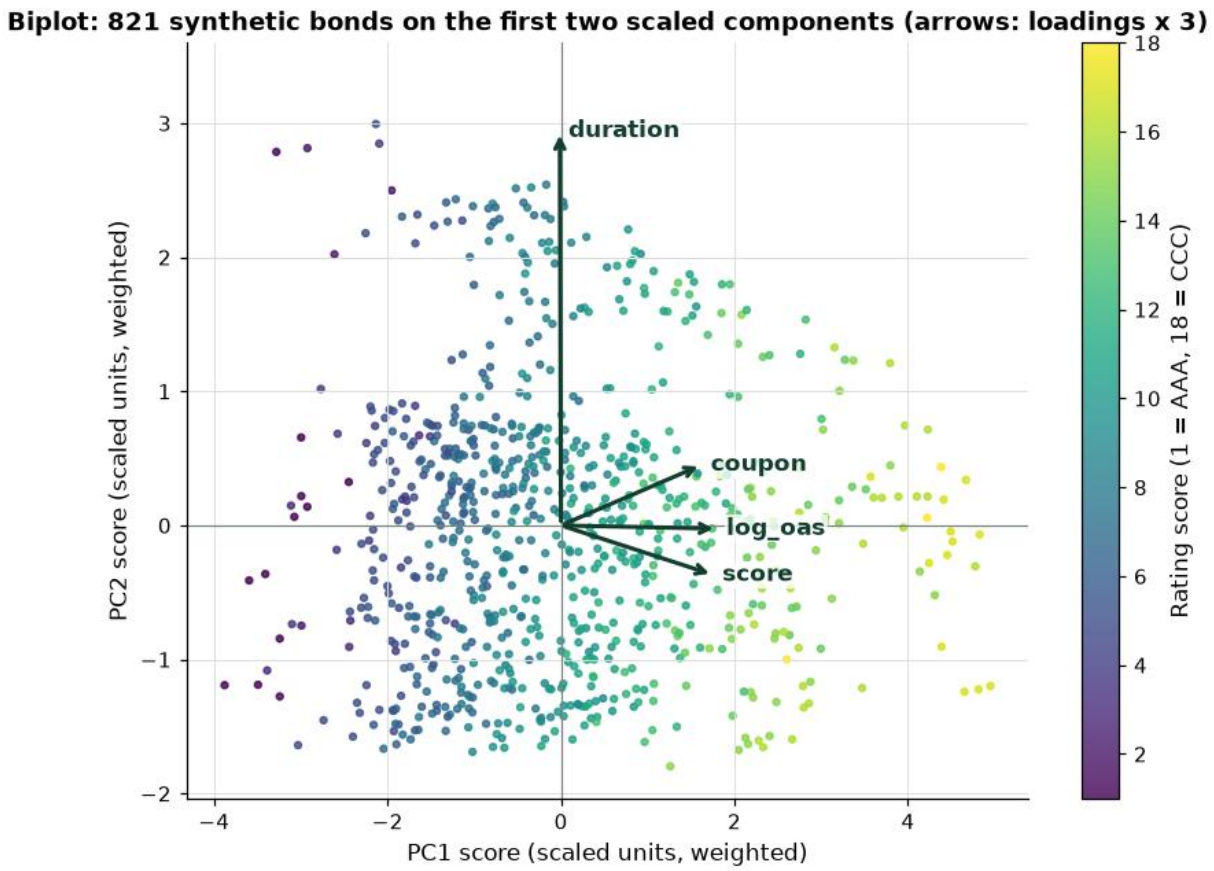


Column	Std, own units	PC1 raw	PC1 scaled
log_oas	0.735	0.179	0.602
duration	3.251	-0.252	-0.005
coupon	1.654	0.317	0.542
score	3.431	0.897	0.586

PC1 carries 54.46 percent raw and 64.96 percent scaled. Unscaled, PC1 is mostly the column with the largest spread in its own units.

Different units: scale first. The bonds are invented.

The biplot: bonds and loadings on one chart



The dots

Each bond's scores on PC1 and PC2, coloured by rating score.

The arrows

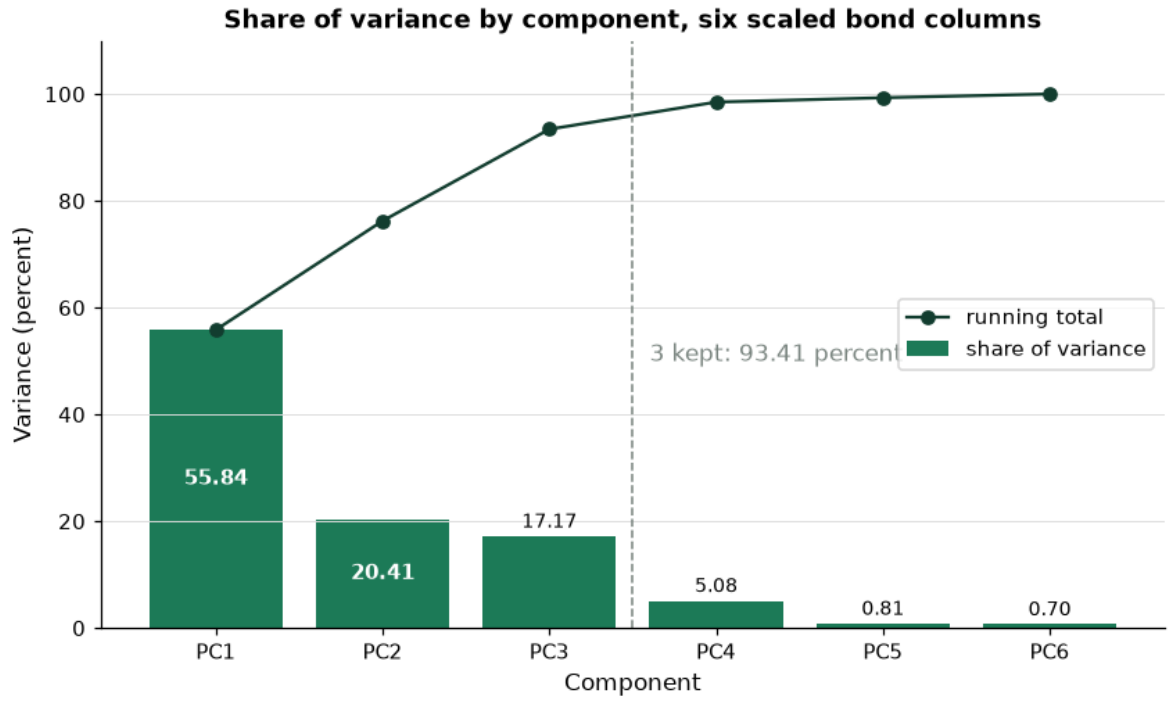
Each column's loadings. Duration points up PC2 (0.981).

The arrows say what each axis is made of.

How a PCA picture reaches five dimensions

Scores	Every bond has a score on each component; the first few carry most of the variation
Three axes	The scores on the first three components are the x, y, and z axes of a 3D scatter
Two channels	A fourth component colours the points, a fifth sets their size
Where it fails	The eye reads position best, colour next, size worst: the scree and loadings still matter
What comes out	A view of the bonds in five directions, charted only: plotly turns, matplotlib stays still

Six columns, up to five dimensions



93.4%

PC1 to PC3: the three axes

+5.08

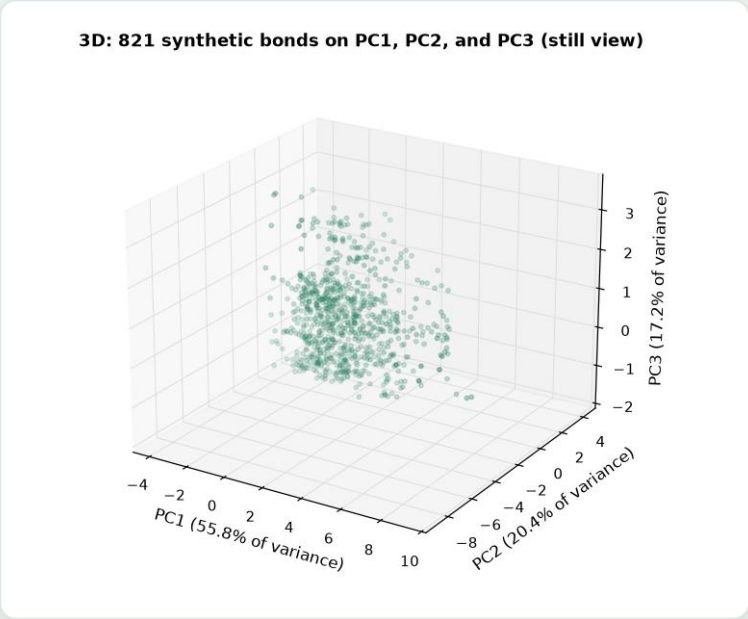
PC4: the colour

+0.81

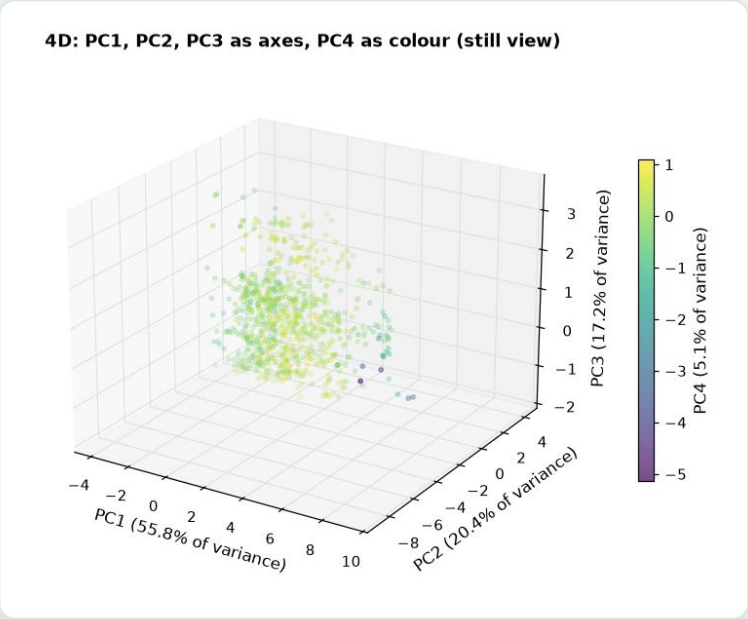
PC5: the size

How much each extra dimension of the picture can show. The yield and the price join the four columns of 14.6.

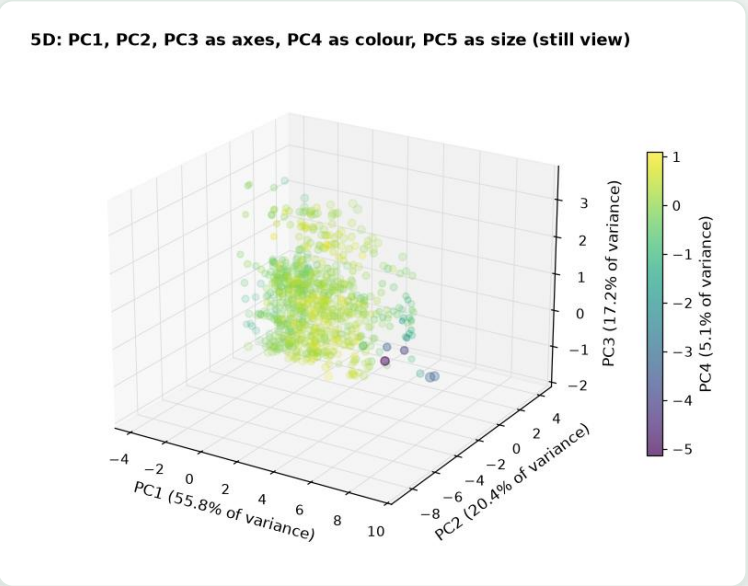
Three, four, five dimensions



3D: PC1 carries 55.84%; three axes show 93.41%.



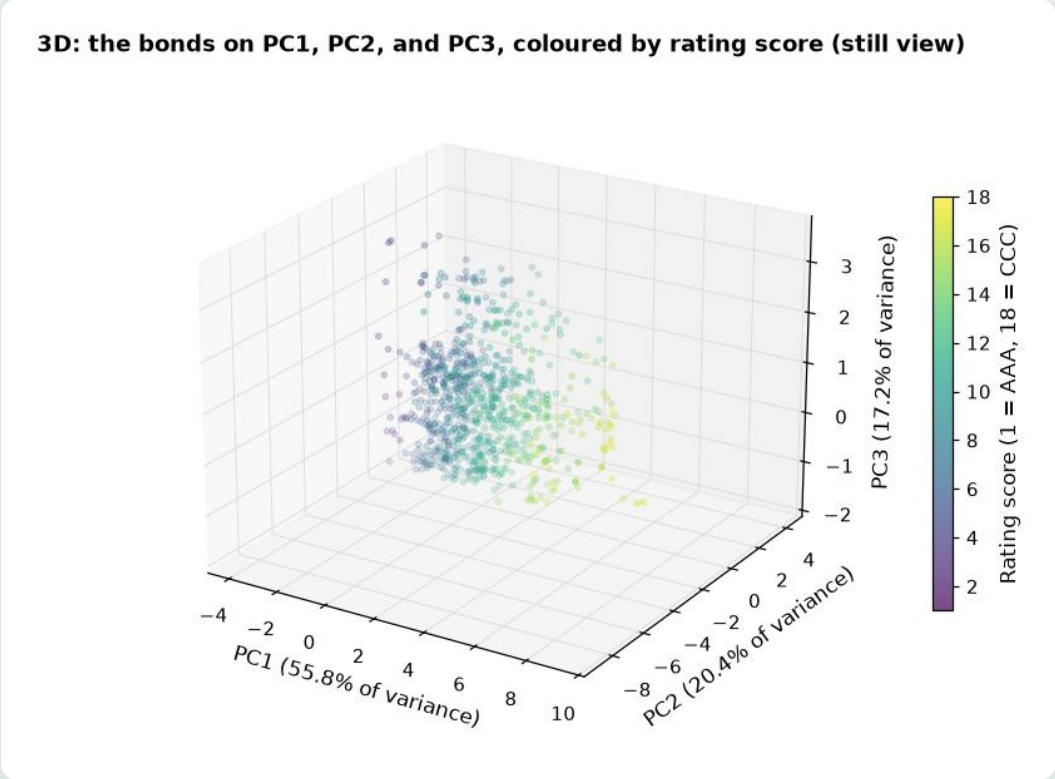
4D: PC4 as colour carries 5.08%: shades are hard to tell apart.

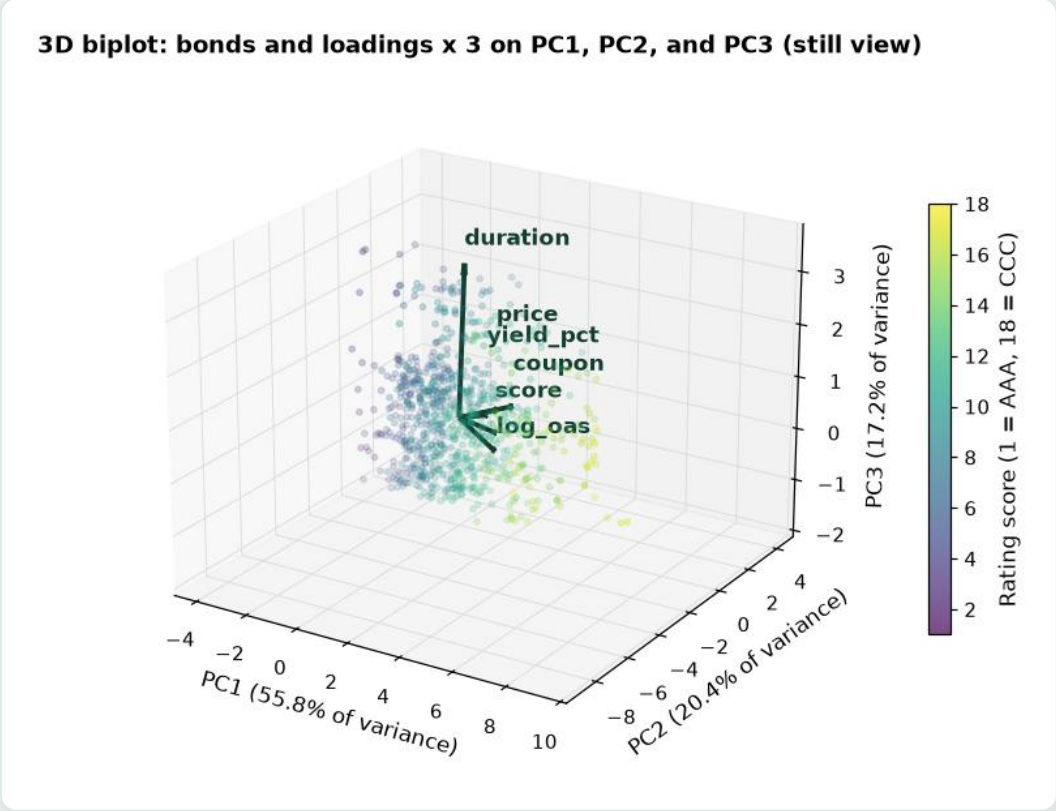


5D: PC5 as size carries 0.81%: the eye fails here.

Each still has a plotly twin in the notebook that turns with the mouse in VS Code and Colab.

By rating score, and the 3D biplot



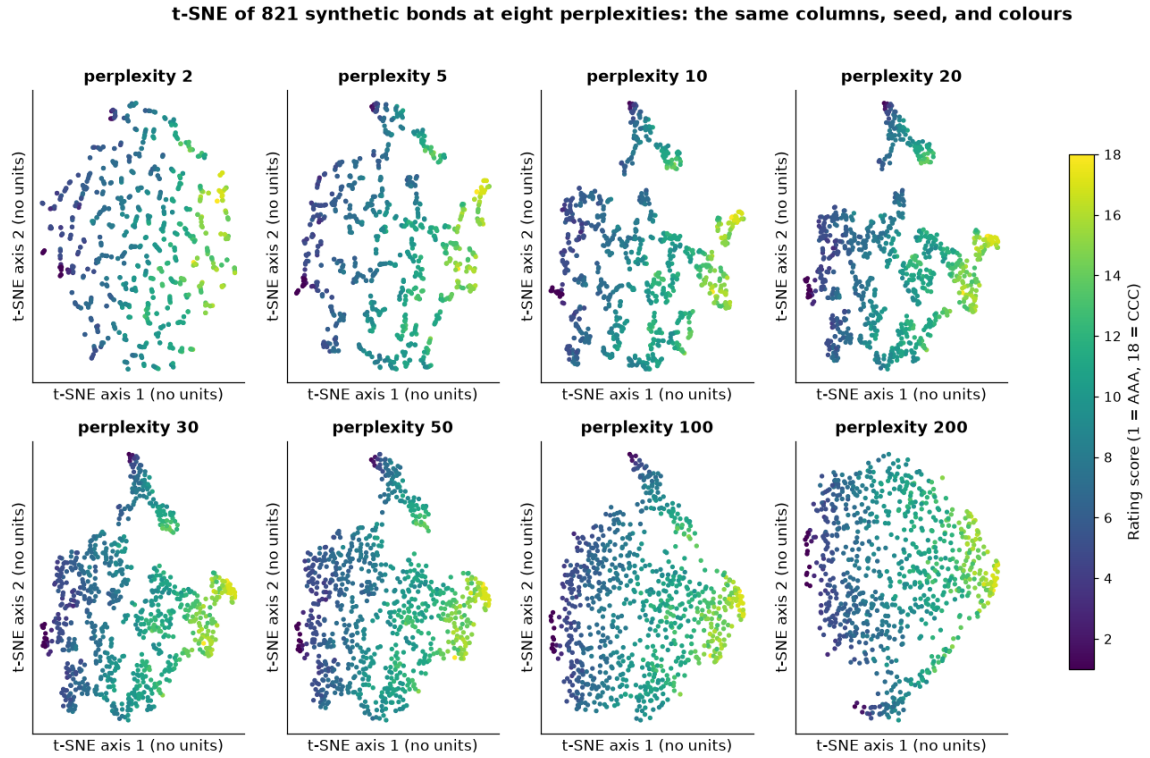


The colour runs along PC1 (correlation 0.92). The arrows say what each axis is made of.

How t-SNE works

The aim	Place each row on a flat page so rows near each other in all the columns stay near each other
Neighbour chances	The distances between rows become the chance that each row picks each other row as a neighbour
The layout	Start from PCA's first two components, scikit-learn's default; move the points until the page's chances match the original ones
You control	perplexity, roughly how many neighbours each point attends to, and the starting layout
What comes out	Two coordinates per row, charted and never printed: only the neighbourhoods mean something

t-SNE: a picture of many columns



Can be trusted for

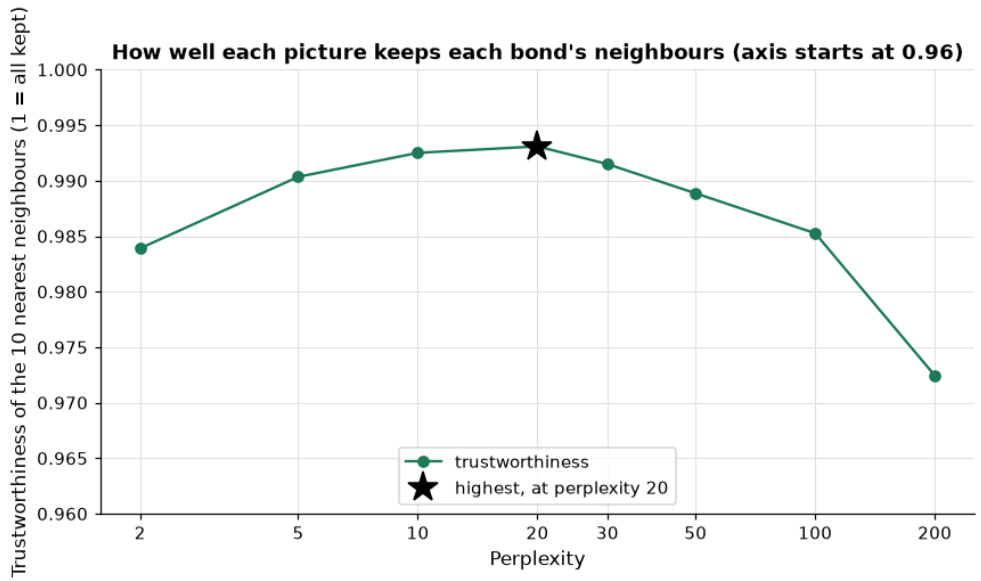
Which rows sit near which; a first look at whether there are groups.

Cannot be trusted for

Distances between groups, group sizes, the axes, anything printed.

One change per panel: the shapes that survive from 10 to 100 are the ones to trust.

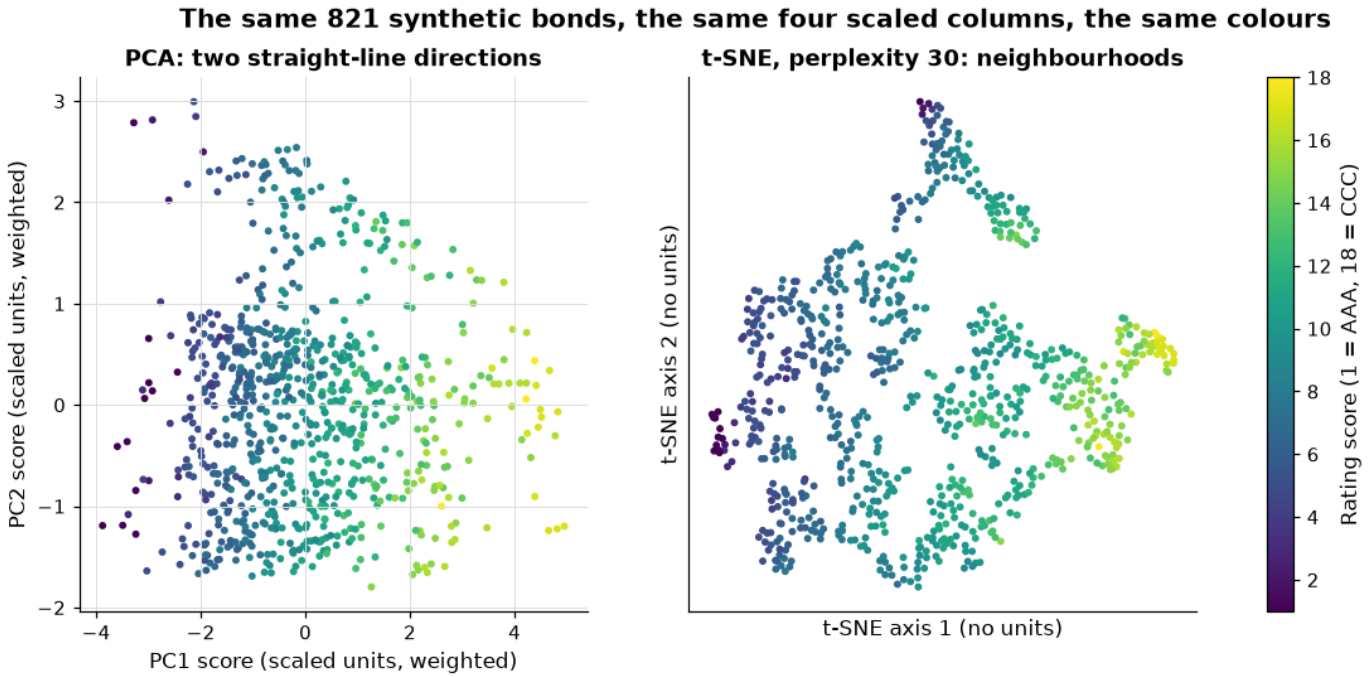
Choosing a perplexity



1. Try a wide range	same data, seed, colours
2. Drop the extremes	noise at the bottom, one cloud at the top
3. Keep what survives	the same shapes panel to panel: 10 to 50
4. Choose inside it	say the value; read neighbourhoods only

scikit-learn's TSNE documentation: Consider selecting a value between 5 and 50. The default is 30.

PCA beside t-SNE, the same colours



PCA's axes are weighted sums you can read. t-SNE's mean nothing: only who sits near whom carries over.

Excel side by side

Excel side by side (tested in Excel)

`=(B3-B2)*100  (K3, fill right and down)`	the changes in bp, exactly
`=CORREL(10 Yr changes, 20 Yr changes)`	0.9529
`=COVARIANCE.S(2 Yr changes, 10 Yr changes)`	20.9902 bp squared
`=MMULT(K3:R2941-BYCOL(K3:R2941,LAMBDA(c,AVERAGE(c))),loadings!B2:D9)`	the scores
`=MMULT(K3:R2941,loadings!B2:D9)`	not the scores: off by 0.333 bp
`=MMULT(covariance, PC1 loadings)/PC1 loadings`	181.4263 on all 8 rows
`=F2/H2, with =SUMPRODUCT(covariance*MUNIT(8)) in H2`	PC1 share 85.04%

Excel has no PCA and no t-SNE. It confirms components it is given; it cannot find them. Centre first.

Save the day: a tag

1. Module

pca_loadings and explained_table, three tests

2. Results

results.csv: the walk-forward pipeline, 4 decimals

3. Tag

git tag week3-teaching

```
git log --oneline --decorate
```

```
HASH (HEAD -> main, tag: week3-teaching) Experiment log: walk-forward test metrics of three curve components then a line
```

A tag names one commit and never moves: the module as it stood at the end of the teaching days.

The AI check

The docstring is the prompt. The code that comes back runs, and has one modelling bug.

1

Docstring

Which rows fit, which rows report, and the units: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: PCA fitted on the whole history, then used for the first fold: 0.8 bp against 0.8 bp. Only the test catches it.

Code that runs is not the same as code that is right.

What you can do now

- Three components: 99.86% of levels, 98.17% of daily changes
- Level, slope, curvature: names for shapes, where the loadings show them
- Scores are the centred data times the loadings; the sign rule fixes the sign
- PCA that builds features goes inside a pipeline; the test proves it
- Scale first on mixed units; t-SNE is charted, never measured

Week 3, Day 5 The capstone: three models on a real file of company ratios